| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 1 | 2 | 4 |
| 2 | 3 | 5 |
| 3 | 4 | 6 |
| 4 | 5 | 7 |

residual ↓ → λ ↑
↑ learned.

Iterat$^n$ 1

$r_1 = y$ a) fit a model $f_1$

$\boxed{x_1 \geq 2.5}$ Decision stump

N → $\boxed{4.5}$ Y → $\boxed{6.5}$

$r^{(1)} = r^{(1)} - \lambda f_1(x^{(1)})$, $\boxed{\lambda = 0.1}$

$= 4 - 0.1(4.5)$

$= 3.55$

$r^{(2)} = 4.55$

$r^{(3)} = 6 - 0.1(6.5)$

$= 5.35$

$r^{(4)} = 7 - 0.1(6.5)$

$= 6.35$

$$r = \begin{bmatrix} 3.55 \\ 4.55 \\ 5.35 \\ 6.35 \end{bmatrix}_{4\times 1}$$

Iteration 2

| $x_1$ | $x_2$ | $r$ |
|---|---|---|
| 1 | 2 | 3.55 |
| 2 | 3 | 4.55 |
| 3 | 4 | 5.35 |
| 4 | 5 | 6.35 |

$\boxed{x_1 \geq 2.5}$

N → $\boxed{4.05}$ Y → $\boxed{5.85}$

$r^{(1)} = 3.55 - 0.1(4.05)$

$= 3.145$

$r^{(2)} = 4.145$

$r^{(3)} = 4.765$, $x^{(4)} = 5.785$

$$r = \begin{bmatrix} 3.145 \\ 4.145 \\ 4.765 \\ 5.765 \end{bmatrix}_{4\times 1}$$

| $x_1$ | $x_2$ | $r$ |
|---|---|---|
| 1 | 2 | 3.145 |
| 2 | 3 | 4.145 |
| 3 | 4 | 4.765 |
| 4 | 5 | 5.765 |

and so on . . .

$$f = \lambda f_1 + \lambda f_2 + \lambda f_3 + \cdots \lambda f_B$$